Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/27/2020
PI Name:	Wegst, Ulrike Ph.D.		
Project Title:	Structure-Property-Processing Correlations in Freeze-Cast Biomimetic Materials		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to Northeastern University from Dartmouth College in summer 2020.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/14/2017	**End Date:**	07/01/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/1/2020 (originally 12/13/2023) due to PI move and continuation grant issued at new institution (Ed., 1/27/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shubitidze, Fridon Ph.D. (Dartmouth College) Karma, Alain Ph.D. (Northeastern University)		
Grant/Contract No.:	80NSSC18K0305		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Strong, tough, and lightweight materials are needed for a myriad of structural and functional applications in Space systems and on Earth in diverse strategic fields that range from biomedical, to buildings, transportation and energy generation, or storage. These new, yet to be developed materials, should exhibit unprecedented combinations of strength and toughness, while being manufacturable at low cost and in high volume. Numerous processing techniques have been proposed to produce these materials, but frequently cannot provide the degree of microstructural control needed to manipulate and optimize the property profile, particularly of multifunctional materials. One technique, which holds great promise in fabricating novel materials with great structural control, is freeze-casting, a relatively inexpensive procedure which also provides a means to mimic complex, efficient natural materials with hierarchical designs over several length-scales. To date, the fundamental science and understanding of this processing technique is limited, because, on Earth, convection and sedimentation complicate both the interpretation of experimental results and their comparison with theoretical predications. Avoiding these in space, following four scientific goals will be pursued. To reveal on Earth (Aim 1) and in Space (Aim 2) the underlying fundamental materials science of freeze casting and establish a realistic model for structure formation in multifaceted crystal nucleation and growth. Aim 3: Develop predictive models to identify the most promising material processing combinations and conditions to optimize material performance. Aim 4: Translate the results of (1-3) into structure-property-processing correlations for highly porous, freeze-cast scaffolds (polymer, ceramic, metal, and composite) made on Earth. Data obtained and computational models will be made available to the NASA Physical Sciences Informatics to make the data accessible to the global community and help translate research into application for an accelerated pace of advanced-materials discovery, innovation, manufacture, and commercialization. NOTE: Grant ended 7/1/2020 (originally 12/13/2023), due to Principal Investigator Ulrike Wegst's move to Northeastern University and new continuation grant 80NSSC21K0039 issued (Ed., 1/27/21)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our goal is to apply the fundamental knowledge gained through this study in the development of new and improved freeze-cast (ice-templated) materials that are strong, tough and lightweight and can be custom-designed for a given application. Such materials are required for applications that range from biomedical and tissue regeneration to filtration, catalysis, transportation, and energy generation and storage. Only with fundamental, physics-based models, systematic materials design and synthesis will this be possible. The results of the terrestrial and microgravity research will be collated, documented, and captured also in a visual format such as in, for example, structure-property-processing charts. With these we wish to enable a paradigm shift towards a more systematic approach in materials synthesis by pointing into particularly promising materials research directions.
	Annual Report, 12/14/2018 – 12/13/2019 Generally, when aqueous solutions or slurries without additives are freeze cast, meaning directionally solidified, the ice crystals grow with their basal plane oriented parallel to the temperature gradient and along the crystallographic <11-20> direction. Because of this orientation, the structure of the freeze-cast wine-cork-like sample is lamellar once a quasi-steady state of solidification has been reached. The first two layers in a sample are dense and cellular in structure, only a few 100 µm thick, and initially form during the first seconds of solidification, with cooling rates so high that no phase separation occurs, but the dissolved or suspended material is trapped within the ice crystals. In some cases, when water-based biopolymer solutions are freeze cast, the final freeze-cast materials and scaffolds exhibit features that are very unusual for materials that have been directionally solidified, for example: i) highly aligned and regularly spaced ridges of uniform height parallel to the direction of solidification, ii) fibrillar pillars and bridges in fiber composites, and iii) fibrillation, a form of self-assembly and structure formation in hierarchically structured biopolymers such as collagen. In this study on “Structure-Property-Processing Correlations in Freeze-Cast Biomimetic Materials,” we are exploring these features not only because they are of interest from a fundamental science point of view, but also because the ridges, pillars, bridges, and fibrillation observed in freeze-cast biopolymer scaffolds can result in highly desirable features for biomedical applications. When, for example, seeding both whole and dissociated embryonic chick dorsal root ganglia (DRGs) on laminin-coated three-dimensional (3D) scaffolds and isolated single lamellae, we observed that the ridges are a very powerful cue that leads to highly aligned neurites within the freeze cast scaffold, which is beneficial for nerve regeneration. While preliminary results showed that a highly aligned porosity and highly aligned and regularly spaced ridges of uniform height can be formed with a range of biopolymers under specific conditions, we found that the fundamental science and underlying principles that govern their formation remain unknown. These principles are critical, particularly for the custom-design of materials for applications that range from the biomedical to energy generation and storage, catalysis, filtration, optical and acoustical absorption, and thermal insulation. We have therefore begun to systematically study the structure-property-processing correlations for ridge formation under well-defined conditions. In the first instance, the structure of chitosan scaffolds frozen with two applied cooling rates was explored both in the fully hydrated and the dry state, and their mechanical properties determined. The results obtained to date for the structural characteristics of the chitosan-water system contribute to a better understanding of the sequence of events during the solidification process that lead to the observed instability formation in the ice phase and the resulting ice templated ridge formation in the chitosan phase. The mechanical properties, such as modulus and yield strength of the freeze-cast chitosan scaffolds decrease with increasing height for both applied cooling rates and will be correlated to the scaffolds structural features such as lamellar spacing (length of the short pore axis), cell wall thickness and other features such as ridges, and preferential molecular alignment with the freezing direction, to explore how these affect the mechanical performance, and to obtain first structure-property-processing correlations. With the structural and mechanical characterization of freeze-cast chitosan scaffolds, we have taken first steps on the path to our long-term goal of this study: to bridge the current critical knowledge gap, which is the lack of understanding of the fundamental mechanisms of structure formation initially in freeze-cast polymers and later also in particle-based ceramic and metal systems, which is due to the unknown thermal properties and freezing behavior of the proposed systems and the kinetics of structure formation. The size and form of the pores in freeze-cast materials are dictated by the intercellular or interdendritic spaces. It is, therefore, critical to develop a fundamental understanding of the 3D arrangement of cells during directional solidification. Because the equilibrium phase diagrams and, even more important, the non-equilibrium state diagrams, which describe

Task Progress:	the metastable ‘states’ fundamental to an understanding of these systems as well as concentration- and temperature-dependent diffusion coefficients and viscosities, are unknown for our water-biopolymer systems, we started by studying in 2D and in 3D the trehalose-water and sucrose-water systems, for which these properties are documented, while determining them for other biopolymer-water systems. Since crystal growth in bulk samples is difficult to observe and characterize in situ, diffusive crystal growth under terrestrial conditions is typically studied with thin samples. These thin samples are designed so that the solidification conditions can be controlled carefully and only a single, two-dimensional (2D) array of dendrites (or cells) forms, and the solid-liquid interface, interface dynamics and the formation of instabilities can be observed. Experimentally, we investigate and determine the kinetics of the ice-phase growth and the mechanisms by which the growing ice-crystals, thus ice phase, templates the polymer phase during its transition from a solution to a glassy solid in water-based system. During the first two years, we have focused on light microscopy perpendicular to the moving freezing front primarily in the trehalose-water and sucrose-water systems, to observe the formation of Mullins-Sekerka instabilities and their subsequent reorganization into an array of cells or dendrites, as a function of polymer concentration and cooling rates, which is equal to the product of velocity and thermal gradient in directional solidification. As expected, the dendrite width in 2D, corresponding to lamellar spacing in 3D, decreases and instability formation increases with increasing freezing front velocity. Overall, our observation is that thin film experiments have to be most carefully controlled to be reproducible and that it is impossible to extrapolate structural properties from thin film to bulk samples, in which also the crystal array formation (e.g., square or hexagonal) have a significant influence on dendrite spacing, sidewise instability formation and the resulting microstructure. Critical is therefore to combine 3D imaging with 3D simulations. The trehalose-water and sucrose-water systems were, in Year 2, the starting point also for the 3D phase-field modelling effort. Using 3D phase-field simulations of directional solidification of water with a dilute impurity (trehalose, sucrose), we reproduced for the first time salient features of freeze-cast structures including, for example, the formation of lamellae and undulated ridges which are particularly clearly visible in purely polymeric systems (e.g., chitosan). The formation of ridges on the polymer lamellae parallel to the direction of solidification could be shown to be caused by the breakdown of ice platelets, which grow in between lamellae in the basal plane of ice, into linear arrays of cells with rounded tips. The cell spacing determines the ridge spacing, and undulations are generated by secondary oscillatory instabilities of cell growth. The value of the 3D phase-field simulations is immense, since they enable us to investigate little studied and not yet modeled mechanisms of hierarchical microstructure formation and cell wall material self-assembly, both of which define anisotropy, property profile, and overall performance of freeze-cast materials. Our results highlight the crucial role of both the solid-liquid interface free-energy and the interface kinetics in structure formation. They also highlight that other potentially important physical effects neglected in the phase-field model, such as fluid flow generated by the expansion of ice, may play a more secondary role in structure formation. From a practical point of view, the ability to accurately predict the detailed temporal evolution of interface structures under prescribed growth conditions is critical for optimizing processing routes and material properties. [Ed. note October 2020: Compiled from PI progress report submitted to NASA Marshall Space Flight Center] ED. NOTE January 2021: Grant ended 7/1/2020 (originally 12/13/2023), due to Principal Investigator Ulrike Wegst's move to Northeastern University and new continuation grant 80NSSC21K0039 issued ; see "Structure-Property-Processing Correlations in Freeze-Cast Biomimetic Materials--80NSSC21K0039" for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 07/29/2025)
Articles in Peer-reviewed Journals	Yin K, Divakar P, Wegst UGK. "Plant-derived nanocellulose as structural and mechanical reinforcement of freeze-cast chitosan scaffolds for biomedical applications." Biomacromolecules 2019 Oct;20(10):3733–45. https://doi.org/10.1021/acs.biomac.9b00784 , Oct-2019
Articles in Peer-reviewed Journals	Yin K, Mylo MD, Speck T, Wegst UGK. "Bamboo-inspired tubular scaffolds with functional gradients." J Mech Behav Biomed Mater. 2020 Oct;110:103826. Epub 2020 May 18. https://doi.org/10.1016/j.jmbbm.2020.103826 ;PMID: 32957175 , Oct-2020
Articles in Peer-reviewed Journals	Yin K, Mylo MD, Speck T, Wegst UGK. "2D and 3D graphical datasets for bamboo-inspired tubular scaffolds with functional gradients: micrographs and tomograms." Data Brief. 2020 Jun 17;31:105870. https://doi.org/10.1016/j.dib.2020.105870 ; PMID: 32642506; PMCID: PMC7334595 , Jun-2020